Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C111 **GDH3** 2 genes / 2 species 1 direct pair	**Y1 GDH3** GenAge 1059 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 20657825	**W1 gdh-1** GenAge 2043 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 22103665	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C042 **GLK1** 4 genes / 2 species 4 direct pairs	**Y1 GLK1** GenAge 1624 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 19030232, 24586198 **Y2 HXK2** GenAge 357 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 11000115, 20421943	No gene in this candidate network	**F1 Hex-A** GenAge 2150 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 26378219 **F2 Hex-C** GenAge 2151 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 26378219	No gene in this candidate network
Direct pairs: F1–Y1 (n:m); F1–Y2 (n:m); F2–Y1 (n:m); F2–Y2 (n:m)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.